

8.1 Supported Communication Protocol Types

4.1.1 Modbus-RTU Communication Protocol (Default from Factory)

Start Bit: 1. Data Bits: 8 Parity Bit: None

Stop Bits: 1 Error Check: CRC (Cyclic Redundancy Check)

Baud Rate: Default from factory is 9600 bit/s

Communication protocol, the format is as follows:

Initial structure with a time of ≥ 4 bytes

Address code = 1 byte Function code = 1 byte Data area = N bytes

Error check = 16-bit CRC code End structure with a time of ≥ 4 bytes

Address code: The address of the transmitter, which is unique in the communication network (default factory setting 0x01).

Function code: The function indication of the command sent by the host, this instrument uses function codes 0x03 and 0x06.

Data area: The data area consists of specific communication data, note that the high byte of 16 bits data is in front.

CRC code: A two-byte check code (low byte first, high byte second).

8.1.2 Example and explanation of the communication protocol

4.1.2.1 If the current temperature value is 100.0, read the temperature value of the instrument at device address 0x01.

Send data 010300000001840A, returned data: 01030203E8B8FA

Send Data

Address Code	Function Code	Starting Address	Checksum Low Byte	Checksum High Byte
0x01	0x03	0x00 0x00	0x84	0x0A

Return Data

Address Code	Function Code	Returned valid byte count	Temperature Data	Checksum Low Byte	Checksum High Byte
0x01	0x03	0x02	0x03E8	0xB8	0xFA

The returned data in signed hexadecimal integer format 0x03E8 converted to decimal is 1000 (the upper computer, PLC, touch screen, and other devices need to divide the data by 10), which is the current temperature value read.

Generally, devices supporting the Modbus-RTU protocol will directly support this, and no additional programming is required to read the data.

8.1.2.2 Method for setting instrument address

The register address is 0x070xA0

Example: Set address 1 to address 2

Send Data

Address Code	Function Code	Address Register	New Address	Checksum Low Byte	Checksum High Byte
0x01	0x06	0x07 0xA0	0x00 0x02	0x09	0x5D

Return Data

Address Code	Function Code	Address Register	New Address	Checksum Low Byte	Checksum High Byte
0x02	0x06	0x07 0xA0	0x00 0x02	0x09	0x6E

8.1.2.3 Instrument address reset (can reset instruments of any address)

Address register is 0x070xA0

Example: Reset the instrument with address 0x0A to address 0x01

Send data (fixed value)

Address Code	Function Code	Address Register	New Address	Checksum Low Byte	Checksum High Byte
0xFA	0x06	0x07 0xA0	0x00 0x01	0x5C	0xD7

Return data (fixed value)

Address Code	Function Code	Address Register	New Address	Checksum Low Byte	Checksum High Byte
0x01	0x06	0x07 0xA0	0x00 0x01	0x49	0x5C

Laser activation method: (This method has no delay function after activating the laser)

Address Code	Function Code	Address Register	Write Data	Checksum Low Byte	Checksum High Byte
0x01	0x06	0x00 0x11	0x11 0x11	0x15	0x93

Laser deactivation method:

Address Code	Function Code	Address Register	Write Data	Checksum Low Byte	Checksum High Byte
0x01	0x06	0x00 0x11	0x00 0x00	0xD9	0xCF

The above communications can be configured through a dedicated computer setup software.